Finance Research Letters 69 (2024) 106293

Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl

Air quality, ES risks, and stock returns: Evidence from Korea

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ARTICLE INFO

JEL Classification:
Q53
G10
G24
G23

Keywords:
Air pollution
Stock return
Behavioral bias
ES risks
Institutional investors

ABSTRACT

This paper examines the impact of air quality index (AQI) on stock returns. Using data from listed Korean firms for the 2015–2021 period, we show that the dispersion in AQI adversely affects stock returns with one-standard-deviation rise in AQI resulting in a 3.0 percentage-point decline in stock returns. This relationship however becomes less pronounced in firms with lower environmental and social (ES) risks and more institutional investments. Our results provide implications for investors and policymakers by highlighting the importance of managing ES risks and the role of institutional investments in mitigating the adverse impact of air pollution on stock returns.

1. Introduction

Air pollution has long been recognized as a critical environmental challenge, contributing significantly to global health issues. Ranked among the top ten contributors to global diseases (OECD, 2012), its effects extend beyond physical health, influencing mental well-being and cognitive functions, which, in turn, can impact economic behaviors and decisions. Studies have shown that exposure to high levels of air pollution correlates with increased rates of depression and anxiety (Vert et al., 2017) and can significantly alter mood and cognitive abilities (Evans et al., 1987; Zeidner and Shechter, 1988; Christelis et al., 2010). These alterations in mood and cognition are critical as they can induce bias in decision-making processes (Loewenstein et al., 2001).

Financial markets are not immune to these influences as market participants rely heavily on cognitive functions and emotional stability. Previous research has documented the effects of air pollution on market dynamics, including stock returns and analysts' forecasting bias. For instance, Levy and Yagil (2011) find that higher air pollution levels negatively affect stock returns in the US market. Similarly, Lepori (2016) reports a significant adverse effect of air pollution on stock returns in Milan, Italy, highlighting how local air quality variations can impact investor behaviors. Li and Peng (2016) provide evidence from China that poor air quality leads to lower stock market performance. Further, Dong et al. (2021) observe that air pollution contributes to forecasting bias among financial analysts in China, indicating that cognitive impairments due to pollution can affect even professional assessments. Li et al. (2021) demonstrate how air pollution could induce a behavioral bias, where investors are more likely to sell winning stocks and hold losing ones. Furthermore, Huang et al. (2020) find that poor air quality can lead to worse investor trading decisions, likely due to its detrimental effects on investor mood and cognitive function. These studies collectively underscore the pervasive influence of air pollution on financial decision-making, offering a foundation for our investigation into the Korean market.

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https://doi.org/10.1016/j.frl.2024.106293
Received 29 April 2024; Received in revised form 5 August 2024; Accepted 11 October 2024
Available online 12 October 2024
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Despite the extensive research on air pollution and financial markets, there is a lack of studies focusing on the firm-level impacts within the Korean context and considering the role of environmental and social (ES) risks and institutional investors. Our study addresses these gaps by examining how air pollution affects stock returns at the firm level in Korea and exploring the moderating effects of ES risks and institutional investment. Utilizing data from Air Korea (https://www.airkorea.or.kr), co-administered by the Korea Environment Corporation and the Ministry of Environment in Korea, this study confirms that dispersion in air pollution adversely affects stock returns. At the same time, the relationship becomes less pronounced when firms have lower environmental and social (ES) risks. Additionally, our results indicate that the presence of institutional investors, who tend to make more rational decisions than individual investors, further mitigates the adverse effects of air pollution on stock returns, considering the role of institutional investors in moderating behavior bias (Boehmer and Kelley, 2009).

This study makes several contributions to the literature. First, it extends the analysis of the impact of air pollution on stock markets to the firm level rather than limiting it to overall market observations. This granular approach enables a detailed examination of how air pollution affects firms differently based on their unique characteristics, helping firms better prepare for the challenges posed by climate change. Second, we broaden the scope of research by considering the role of environmental, social, and governance (ESG) ratings. Our findings suggest that ES risks measured as the firms' ES ratings influence investor mood and mitigate the behavior bias induced by air pollution. As air quality worsens, firms with lower ES risks appear more attractive to investors, potentially leading to superior returns than those of the broader market. This phenomenon reverses the typical negative relationship observed between air pollution and stock performance, as documented in prior studies (e.g., Levy and Yagil, 2011; Lepori, 2016; Li and Peng, 2016; Dong et al., 2021; Li et al., 2021). Third, this research contributes a novel perspective by suggesting that variations in institutional trading activity could be a critical determinant in mitigating the behavioral bias induced by air pollution. This insight adds a new dimension to our understanding of how institutional behavior might shield markets from the cognitive distortions caused by environmental stressors.

The remainder of the paper is organized as follows. Section 2 develops the testable hypotheses. Section 3 introduces the study's data and variables. Section 4 explains the empirical results and documents the heterogeneity effect. Section 5 concludes.

2. Literature review and hypothesis development

This section develops our research hypotheses discussing the impact of air pollution on the stock market. The detrimental effects of air pollution on human health are well-documented, with severe levels linked to increased incidences of depression, anxiety, and impaired cognitive functions (Evans et al., 1987; Zeidner and Shechter, 1988; Vert et al., 2017). These impairments are particularly concerning for financial decision-making, where cognitive ability and emotional stability play crucial roles. Lucey and Dowling (2005) highlight how emotions influence investor decisions, while Loewenstein et al. (2001) discuss how mood and cognition induced by external factors such as air pollution can lead to significant decision-making bias. Kamstra et al. (2003) also propose a type of depression that occurs at a specific time of year, primarily in the winter, affects investors' risk aversion and consequently influences stock market behavior. Building on this, we propose our first hypothesis:

H1. There is a negative relationship between the level of air pollution and a firm's stock return.

As air quality deteriorates, environmental concerns gain prominence among investors. Investors are increasingly attentive to the environmental impact of their investment choices, and this growing awareness influences stock prices (Bolton and Kacperczyk, 2021). Notably, when firms are reported to engage in ESG-friendly initiatives, they tend to attract more attention from investors than those not featured in such news, as Barber and Odean (2007) outlined. Consequently, as air pollution levels rise, investors will increasingly seek firms with robust ESG ratings (Jo et al., 2022). This heightened attention will likely result in positive market responses, notably higher stock returns. In addition, investors care about non-pecuniary motives as well, further increasing the demand for stocks with higher ESG ratings (Hong and Kacperczyk, 2009; Dasgupta et al., 2023; He et al., 2023). Specifically, because air pollution is more closely related to environmental and socially responsible activities, we focus on ES ratings in this paper and use the ratings to measure the presence of ES risks (Hübel and Scholz, 2019). Thus, our second testable hypothesis is formulated as follows:

H2. Low ES risks mitigate the adverse impact of air pollution on stock returns.

Individual investors are found to be more biased due to their decision-making based on moods or sentiments (Chuang and Susmel, 2011; Huang et al., 2020). Thus, the adverse impact of air pollution on stock returns will be more pronounced for individual investors. On the other hand, institutional investors are known for their rational and information-based decision-making and access to private information, increasing market efficiency (Boehmer and Kelley, 2009; Edmans, 2009; Kim et al., 2019; Kim et al., 2019). Thus, their trading behaviors could reduce the adverse effects of air pollution-induced behavior bias on stock returns. Hence, our third hypothesis is:

H3. The negative impact of air pollution on stock returns is less pronounced on days with higher institutional trading activities.

In the following sections, we introduce the data and variables used in our study, followed by a detailed discussion of our empirical results and their implications.

3. Data and methodology

3.1. Data description

The raw Air Quality Index (AQI) at the provincial-daily level is collected from Air Korea (https://www.airkorea.or.kr), co-administered by the Korea Environment Corporation and the Ministry of Environment in Korea. AQI measures the level of the air pollutant with the highest concentration in the air among six significant pollutants: sulfur dioxide (SO_2), carbon monoxide (CO), ozone (O_3), nitrogen dioxide (NO_2), particulate matter with diameters less than 10 micrometers (PM10), and particulate matter with diameters less than 2.5 micrometers (PM2.5).[1] To account for seasonality in the air quality data, we apply a seasonality adjustment to the AQI by subtracting the quarterly average AQI from the raw AQI.[2]

The stock market- and firm-level accounting data are obtained from the FnGuide database. Our sample covers 1708 days and 931 firms in the 15 provinces from 2015 to 2021. We also collect the daily weather conditions, such as precipitation, humidity, clouds, speed of wind, and temperature from the Korea Meteorological Administration. To avoid the potential effect of outliers, we winsorize all continuous variables by the top and bottom 1 %. Firms in financial or utility industries are excluded from the sample, and observations with missing firm-level controls are eliminated. To be included in the sample, a firm must have positive total assets and disclose ESG ratings evaluated by the Korea Institute of Corporate Governance and Sustainability. Table 1 defines the variables used in this paper, and Table 2 summarizes the descriptive statistics of these variables. The mean value of *RET* is −0.009, with a standard deviation of 2.354, indicating no substantial difference between individual stock returns and market returns. The mean value of *AQI_SA* is −0.004, indicating that the original air quality is slightly better than the seasonal average. The mean value of *Low ES Risk* is 0.156, indicating that 15.6 % of firms have ES ratings higher than "B+," and the mean value of *High_Inst*, 0.838, indicates that institutional involvement is more significant than individual investors' involvement in price decisions on 83.8 % of the observed days.

3.2. Empirical method

To examine the impact of air pollution on the stock returns, we use the following Eq. (1) as the baseline model of estimation:

$$RET_{i,j,t} = \alpha + \beta AQI_SA_{j,t} + \gamma X_{j,t} + \mu Y_{i,t-1} + Firm\ FE + Time\ FE + \varepsilon_{i,j,t} \tag{1}$$

where $RET_{i,j,t}$ represents the market-adjusted stock return of a firm i located at province j on date t. $AQI_SA_{j,t}$ is the daily value of the seasonality-adjusted air quality index at province j, scaled by 100 for the ease of interpretation of the regression coefficients. $X_{j,t}$ is a vector of province-level weather-related control variables. $Y_{i,t-1}$ is a vector of lagged firm-related characteristics.[3] Firm, year-by-month, and day-of-the-week fixed effects are also included in the model specification to absorb time-trend variations and control for all unobserved firm-level heterogeneity that does not vary over time. The coefficient of interest, β, captures the estimated effect of dispersion in air quality on stock returns.

4. Empirical results

4.1. Baseline analysis

Table 3 presents the baseline results regarding the effect of air pollution on stock return using Eq. (1). Column 1 includes year-by-month and day-of-the-week fixed effects. Industry and firm fixed effects are added in columns 2 and 3, respectively. All columns report the negative and statistically significant coefficients for *AQI_SA* at the 1 % level, indicating that the higher dispersion in air pollution is associated with a lower market-adjusted stock return. This effect is economically significant as a one-standard-deviation increase in AQI results in a 3.00 % point decline[4] in stock returns annually. These results are consistent with *H1*.[5] The coefficients of the weather controls are consistent with prior studies (Saunders, 1993; Keef and Roush, 2002; Hirshleifer and Shumway, 2003).

[1] To address the situations where the level of more than one pollutant may exceed certain levels that threaten human health, specific values are additionally assigned to AQI depending on the number of pollutants with significant levels in the air. See 'What's CAI' at Air Korea for a detailed definition of AQI.

[2] The effect of air pollution is most severe shortly after exposure begins but subsequently diminishes (Weitekamp and Hofmann, 2021), so we measure the relative severity of air pollution compared to the seasonality of air pollution in Korea (Yi et al., 2010).

[3] To control for additional confounding factors that may affect stock return, we include firm-level variables, such as size (Banz, 1981), book-to-market ratio (Rosenberg et al., 1985), leverage (Bhandari, 1989), intangible assets (Dong et al., 2021), and momentums (Cremers and Pareek, 2015), and weather-related variables that are known to affect human moods, such as clouds, wind, precipitation, humidity, and temperature (Saunders, 1993; Keef and Roush, 2002; Hirshleifer and Shumway, 2003; Cao and Wei, 2005)."

[4] The calculation for this figure is as follows: 3.00%p = 250 days × −0.0383 × 0.313

[5] To mitigate reverse causality or omitted variable concerns, we perform an instrument variable (IV) regression using the air quality of China as an instrument. The Chinese Air Quality Index (AQI) is also addressed its seasonality by applying the same adjustment to the Korean AQI. We use one-day lagged Chinese AQI because it takes approximately twelve hours for air pollution from China to travel to Korea (Crawford et al., 2021). The (untabulated) result of the first-stage regression is positive and statistically significant at the 1% level, and the coefficient of the estimated Korean AQI from the second-stage regression is negative and statistically significant at the 1% level, consistent with the result of the baseline model.

Table 1
Variable definitions.

Variables	Definitions
Dependent variable	
RET	(The difference between the daily stock return of a firm *i* at date *t* and the daily market return at date *t*) x 100
Independent variable	
AQI_SA	The difference between the daily average air quality index and the quarterly average air quality index at the province level (Seasonally-adjusted)
Mediating variables	
Low ES Risk	An indicator variable which equals one if one of a firm's environmental and social (ES) ratings is above "B+" ('A' or 'A+'), and zero otherwise
Low E Risk	An indicator variable which equals one if a firm's environmental (E) rating is above "B+," and zero otherwise
Low S Risk	An indicator variable which equals one if a firm's social (S) rating is above "B+," and zero otherwise
Inst_IMB_abs	The absolute value of the ratio of institutional investors' net trading volumes (buying volumes – selling volumes) to institutional investors' total trading volumes (buying volumes + selling volumes) of firm i at day *t*
Ind_IMB_abs	The absolute value of the ratio of individual investors' net trading volumes (buying volumes – selling volumes) to individual investors' total trading volumes (buying volumes + selling volumes) of firm *i* at day *t*
High_Inst	An indicator variable which equals one if the absolute value of institutional investors trading imbalance of a firm (*Inst_IMB_abs*) is greater than the one of individual investors' trading imbalance (*Ind_IMB_abs*), and zero otherwise
Weather controls	
Cloud_SA	The difference between the daily average level of clouds and the quarterly average level of clouds at the province level (Seasonally-adjusted)
Wind Speed_SA	The difference between the daily average wind speed and the quarterly average wind speed at the province level (Seasonally-adjusted)
Precipitation_SA	The difference between the daily average level of precipitation and the quarterly average level of precipitation at the province level (Seasonally-adjusted)
Humidity_SA	The difference between the daily average level of humidity and the quarterly average level of humidity at the province level (Seasonally-adjusted)
Temperature_SA	The difference between the daily average temperature and the quarterly average temperature at the province level (Seasonally-adjusted)
Firm controls	
Size	Natural logarithm of total assets at the end of the previous year
Book-to-market	The ratio of book value to market value at the end of the previous year
Leverage	The ratio of liabilities to total assets at the end of the previous year
Intangible Asset	The ratio of intangible assets to total assets at the end of the previous year
Cash Flow	The ratio of net cash flow from operating activities to total assets at the end of the previous year
Momentum 1m	(The difference between the stock return of the previous month and the market return of the previous month at date *t*) x 100
Momentum 12m	(The difference between the stock return of the previous year and the market return of the previous year at date *t*) x 100

This table provides the descriptions of all the variables examined in our analyses.

Table 2
Summary statistics.

Variables	Obs.	Mean	SD	Min	p25	Median	p75	Max
RET	1,149,317	−0.009	2.354	−6.376	−1.289	−0.166	1.033	8.885
AQI_SA	1,149,317	−0.004	0.313	−0.511	−0.205	−0.065	0.119	1.148
Low ES Risk	1,149,317	0.156	0.363	0	0	0	0	1
Low E Risk	1,149,317	0.084	0.277	0	0	0	0	1
Low S Risk	1,149,317	0.140	0.347	0	0	0	0	1
Inst_IMB_abs	1,149,317	0.548	0.352	−0.818	0.218	0.527	0.943	1
Ind_IMB_abs	1,149,317	0.136	0.148	−1	0.031	0.084	0.189	1
High_Inst	1,149,317	0.838	0.368	0	1	1	1	1
Cloud_SA	1,149,317	−0.050	2.846	−6.891	−2.420	−0.076	2.280	6.964
Wind Speed_SA	1,149,317	0.019	0.709	−2.871	−0.457	−0.140	0.346	9.421
Precipitation_SA	1,149,317	−1.284	10.845	−29.112	−4.861	−2.499	0	251.89
Humidity_SA	1,149,317	−0.315	12.639	−50.603	−8.911	−0.606	7.801	46.538
Temperature_SA	1,149,317	−0.037	5.258	−18.214	−3.527	0.236	3.729	16.250
Size	1,149,317	20.317	1.559	15.777	19.214	20.079	21.146	24.659
Book-to-market	1,149,317	1.298	1.108	0.068	0.523	1.036	1.74	5.4
Leverage	1,149,317	0.436	0.201	0.015	0.270	0.439	0.587	1.467
Intangible Asset	1,149,317	0.534	0.173	0.112	0.415	0.536	0.656	0.903
Cash Flow	1,149,317	0.054	0.071	−0.174	0.016	0.053	0.092	0.246
Momentum 1m	1,149,317	0.668	12.871	−30.671	−6.560	−0.611	6.088	52.470
Momentum 12m	1,149,317	−0.044	1.194	−1.628	−0.059	−0.048	−0.031	382.870

This table reports the summary statistics of all the variables used in our study. All the variables are defined in Table 1.

4.2. Effect of ES risks

This section examines how the ES ratings affect the relationship between air pollution and stock returns. To do so, we employ Eq. (2) as follows:

$$RET_{i,j,t} = \alpha + \beta_1 AQI_SA_{j,t} + \beta_2 Low\ ES\ Risk_{i,t-1} + \beta_3 AQI_SA_{j,t} \times Low\ ES\ Risk_{i,t-1} + \gamma X_{j,t} + \mu Y_{i,t-1} + Firm\ FE + Time\ FE + \varepsilon_{i,j,t} \quad (2)$$

Table 3
Baseline regression analysis.

VARIABLES	(1) *RET*	(2) *RET*	(3) *RET*
AQI_SA	−0.0383***	−0.0383***	−0.0384***
	(−3.43)	(−3.43)	(−3.17)
Cloud_SA	−0.0024**	−0.0024**	−0.0024**
	(−2.56)	(−2.56)	(−2.59)
Wind Speed_SA	−0.0256***	−0.0256***	−0.0256***
	(−9.10)	(−9.10)	(−9.18)
Precipitation_SA	−0.0000	−0.000	0–0.0000
	(−0.04)	(−0.04)	(−0.05)
Humidity_SA	−0.0000	−0.0000	−0.0000
	(−0.03)	(−0.03)	(−0.03)
Temperature_SA	0.0054***	0.0054***	0.0054***
	(3.40)	(3.40)	(3.40)
Size	0.0058***	0.0058***	−0.0214***
	(8.34)	(6.50)	(−3.04)
Book-to-market	0.0023***	0.0037***	0.0201***
	(3.10)	(5.25)	(9.65)
Leverage	−0.0195**	−0.0170*	0.0411**
	(−2.59)	(−2.01)	(2.45)
Intangible Asset	−0.0017	0.0011	−0.0226
	(−0.36)	(0.19)	(−1.45)
Cashflow	0.0514***	0.0527***	−0.0125
	(4.59)	(4,93)	(−0.80)
Momentum 1m	0.0364***	0.0364***	0.0363***
	(123.33)	(123.76)	(118.13)
Momentum 12m	−0.0955***	−0.1034***	−0.0559
	(−11.50)	(−5.24)	(−1.01)
Observations	1,149,317	1,149,317	1,149,317
Year*Month FE	YES	YES	YES
Day-of-the-week FE	YES	YES	YES
Industry FE	NO	YES	NO
Firm FE	NO	NO	YES
Adjusted R^2	0.0373	0.0373	0.0368

This table presents the baseline regression restuls from Eq (1) for our main sample between 2015 and 2021. The dependent variable is the market-adjusted stock return (*RET*), and the main independent variable is the seasonally-adjusted AQI (*AQI_SA*). All other variables are defined in Table 1. In column 1, we include year-by-month and day-of-the-week fixed effects in our analysis. In columns 2 and 3, we add industry and firm fixed effects, respectively. All continuous variables are winsorized at 1 % and 99 % levels. The t-statistics reported in parentheses are based on standard errors clustered at the province level. *, **, and *** denote significance at the 10 %, 5 %, and 1 % levels, respectively.

where $Low\ \ ES\ \ Risk_{i,t-1}$ is an indicator that takes a value of one if the ES rating of a firm i is above the threshold, "B+." The interaction term between the seasonality-adjusted air quality index and the presence of low ES risks is added in Eq (2).[6] The coefficient on the interaction term, β_3, measures the mitigating effect of ES ratings on air pollution.

Table 4 reveals the results of Eq. (2). Column 1 includes the interaction term between seasonally-adjusted air quality and overall ES risks. While the coefficient of *AQI_SA* is negative and statistically significant, the coefficient of the interaction term is positive and statistically significant at the 5 % level. Consistent with our conjecture that increasing attention on ES risks leads to positive market reactions, the low ES risk of a firm mitigates the adverse impact of air pollution on stock returns. This mitigating effect is also economically significant as firms with lower ES risks experience only a 0.38 percentage point decline in stock returns annually, while firms with higher ES risks experience a decline of 3.48 percentage point in stock returns annually. Column 2 includes the interaction between air pollution and low environmental (E) risks, and column 3 involves the interaction term with low social (S) risks. Both columns have consistent results with the results in column 1, which confirms *H2*.[7]

[6] Data on the ESG ratings of firms in the KSE and the KOSDAQ markets is obtained from the Korea Exchange (KRX) database. The KRX receives the ESG rating data evaluated by the Korea Institute of Corporate Governance and Sustainability (KCGS) each year and discloses the information to the public. The ratings are divided into six levels, denoted as A+, A, B+, B, C, and D, and firms disclose their ratings on the overall ESG, environmental, social, and governance sections. Though the disclosure is not mandatory in Korea yet, an increasing trend has been shown for firms disclosing their information.

[7] In untabulated results, we conduct subsample analyses dividing the sample into firms with high environmental and/or social risks versus those with low environmental and/or social risks. Consistent with the results in Table 4, the effect of air quality level on stock returns is statistically significant only in firms with high environmental and/or social risks. We thank the anonymous referee for pointing out that this specific study would migate the concern of multi-collinearity across different dummy variables.

Table 4
Impact of ES risks on market-adjusted stock returns.

VARIABLES	(1) *RET*	(2) *RET*	(3) *RET*
AQI_SA	−0.0445***	−0.0427***	−0.0450***
	(−4.20)	(−4.08)	(−4.18)
*AQ_SA*Low ES Risk*	0.0418**		
	(2.52)		
Low ES Risk	−0.0021		
	(−1.12)		
*AQ_SA*Low E Risk*		0.0519*	
		(2.10)	
Low E Risk		0.0020	
		(0.71)	
*AQI_SA*Low S Risk*			0.0473**
			(2.42)
Low S Risk			−0.0004
			(−0.22)
Observations	1,149,317	1,149,317	1,149,317
Weather Controls	YES	YES	YES
Firm Controls	YES	YES	YES
Year*Month FE	YES	YES	YES
Day-of-the-week FE	YES	YES	YES
Firm FE	YES	YES	YES
Adjusted R^2	0.0368	0.0368	0.0368

This table reports the OLS regression results for the interactions between the seasonally-adjusted air quality and the environmental and social (ES) indicators on market-adjusted stock returns. In column 1, we add the interaction with the overall ES indicator, which is equal to one if the environmental or the social rating of a firm is above "B+", and in column 2 and 3, we add the interaction with the E and the S indicators, respectively. The dependent variable is the market-adjusted stock return (*RET*), and the main independent variable is the seasonally-adjusted AQI (*AQI_SA*). All other variables are defined in Table 1, and all continuous variables are winsorized at 1 % and 99 % levels. The t-statistics reported in parentheses are based on standard errors clustered at the province level. All specifications include firm, year-by-month, and day-of-the-week fixed effects, and all firm-related controls are lagged by one year. The sample period is from 2015 to 2021. *, **, and *** denote significance at the 10 %, 5 %, and 1 % levels, respectively.

4.3. Mechanism: Institutional trading activity

This section investigates how air pollution-induced behavioral bias is differentiated by different types of investors' trading activities. Specifically, we proxy investors' trading activities as trading imbalances. Obrimah and Wong (2022) theoretically explain how trading imbalances affect stock returns in the continuous time framework, and Obrimah (2023) also presents that trading imbalance affects stock returns through swamping information. We measure institutional (individual) investor's trading imbalance, "*Inst_IMB_abs*" ("*Ind_IMB_abs*") as the absolute value of the ratio of institutional (individual) investors' net trading volumes to total trading (Griffin et al., 2003; Kim and Stoll, 2014). The ratios capture how much each investor contributes to market incompleteness through excess supply or demand, eventually affecting stock returns. For example, if *Inst_IMB_abs* is close to one, the institutional investors have an impact on prices compared to the equilibrium of supply or demand through swamping information (Stoll, 2000). Columns 1, 2, and 3 in Table 5 include days when *Inst_IMB_abs* is less than *Ind_IMB_abs* (i.e., *High_Inst* = 0), whereas columns 4, 5, and 6 include days when *Inst_IMB_abs* is greater than *Ind_IMB_abs* (i.e., *High_Inst* = 1). Consistent with our conjecture, when swamping information from individual investors affects stock prices more than institutional investors (columns 1, 2, and 3), the coefficients of the interaction terms between *AQI_SA* and ES-related indicator variables are statistically significant at the 1 % level and have greater magnitudes than when institutional investors have more impact (columns 4, 5 and 6). Column 7 presents that the differences between the coefficients are statistically significant at the 1 % level, which confirms *H3*.

5. Conclusion

This study empirically examines the relationship between the dispersion in air quality and stock returns in the Korean stock market. Across various classifications with controls and fixed effects, we find a negative and significant relationship between dispersion in air quality and market-adjusted stock returns. We also document that the effect of air pollution is mitigated and even reversed when firms have lower ES risks. Moreover, we demonstrate that the significant and adverse effects of air pollution on stock returns are evident for firms on days with lower levels of institutional investments. This study extends the existing literature on the relationship between air pollution and stock returns and broadens the role of ES risks in behavior bias. Finally, our results suggest that lower ES risks can attenuate the behavioral bias of air pollution on stock returns.

Author statement

We confirm that our paper is not under consideration for publication elsewhere, that its publication is approved by all authors and

Table 5
Heterogeneity effect: Institutional versus individual trading imbalance.

	High_Inst=0			*High_Inst*=1			
VARIABLES	(1) *RET*	(2) *RET*	(3) *RET*	(4) *RET*	(5) *RET*	(6) *RET*	(7) *Low - High*
AQI_SA	−0.0486***	−0.0484***	−0.0525***	−0.0445***	−0.0419***	−0.0440***	
	(−3.37)	(−4.64)	(−4.04)	(−3.73)	(−3.53)	(−3.66)	
*AQI_SA*Low ES Risk*	0.0687**			0.0307*			0.0380***
	(2.84)			(1.88)			(0.008)
Low ES Risk	−0.0231			0.0137***			
	(−1.56)			(6.03)			
*AQI_SA*Low E Risk*		0.1203***			0.0196		0.1007***
		(5.65)			(0.67)		(0.00)
Low E Risk		−0.0346**			0.0253***		
		(−2.57)			(6.30)		
*AQI_SA*Low S Risk*			0.0917***			0.0297	0.0619***
			(4.11)			(1.41)	(0.00)
Low S Risk			−0.0157**			0.0167**	
			(−2.48)			(2.58)	
Observations	186,111	186,111	186,111	963,199	963,199	963,199	
Weather Controls	YES	YES	YES	YES	YES	YES	
Firm Controls	YES	YES	YES	YES	YES	YES	
Year*Month FE	YES	YES	YES	YES	YES	YES	
Day-of-the-week FE	YES	YES	YES	YES	YES	YES	
Firm FE	YES	YES	YES	YES	YES	YES	
Adjusted R^2	0.0363	0.0363	0.0363	0.0375	0.0375	0.0375	

This table presents the regression results using investors' trading imbalance as a mediating variable. We add the interaction terms between the air quality (*AQI_SA*) and the ES indicators as in Table 4 and examine the role of investors' trading imbalances by separately examining firms with low vs. high institutional investments relative to individual investments. Columns 1, 2, and 3 represent the days with low institutional investments (*High_Inst* = 0), and columns 4, 5, and 6 represent the days with high institutional investments (*High_Inst* = 1). Column 7 presents the results of the mean difference tests between the coefficients of the interaction terms in columns 1 and 4, columns 2 and 5, and columns 3 and 6, respectively. The dependent variable is the market-adjusted stock return (*RET*), and the main independent variable is the seasonally-adjusted AQI (*AQI_SA*). All other variables are defined in Table 1, and all continuous variables are winsorized at 1 % and 99 % levels. The t-statistics reported in parentheses are based on standard errors clustered at the province level. All specifications include firm, year-by-month, and day-of-the-week fixed effects, and all firm-related controls are lagged by one year. The sample period is from 2015 to 2021. *, **, and *** denote significance at the 10 %, 5 %, and 1 % levels, respectively.

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CRediT authorship contribution statement

Byunghun Han: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Junho Park:** Writing – review & editing. **Kwangwoo Park:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization.

Data availability

Data will be made available on request.

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